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Nasalization in the Sara Languages

by Brent H. Vine

0. Preliminaries

The Sara languages are spoken in the southwestern corner of Chad; grammatical descriptions and lexical information on the principal languages consist of the following:

Sara 'proper' (Majingay, lit. 'very-good', dialect): Hallaire and Robinne 1959 (extensive lexical material (tones generally marked), with phonological and grammatical sketch).

Mbay (NB: does not equal Tucker and Bryan's 'Sara Mbai', which corresponds to 'Sara proper'): Fortier 1961 (fairly extensive lexical material (tones only sparsely marked), with phonological and grammatical sketch); — Caprile 1968 (phonological description, including tone, of the southernmost area of the dialect covered by Fortier).

Ngambay (often referred to as 'Sara-Ngambay'): Vandame 1963 (extensive grammatical description and texts, with short glossary); — Thayer and Thayer 1971 (henceforth simply 'Thayer 1971'; learner's

manual: volumes I and II, texts (397 pp.), with translations and grammatical notes, and volume III, phonological and grammatical description (70 pp.), plus glossary). — Both Vandame and Thayer describe the Moundou dialect.

In what follows, these languages will be abbreviated as 'Sa.', 'Mb.' (Dict. refers to Fortier 1961, Cap. to Caprile 1968), 'Ng.' (V. refers to Vandame 1963, Th. to Thayer 1971).

For an informal sociolinguistic description of the area, including some remarks on mutual intelligibility, see Caprile 1968:41ff.

Together with Bagirmi, the Sara languages form the Western branch of the so-called Bongo-Bagirmi languages. (For basic grammatical information on these languages, see Tucker and Bryan 1966:59–83; for geographical/dialectal information see Tucker and Bryan 1956:10ff. The position of the Sara languages within Bongo-Bagirmi, as well as the genetic unity of the Bongo-Bagirmi group as a whole, are immediately apparent upon inspection of the available lexical and grammatical material. The remoter affinities of this group will not be of interest in the discussion which follows.)

For Bagirmi, comparative lexical forms have been drawn from Gaden 1909, whose material tallies closely, on the whole, with the more recent Bagirmi material in Tucker and Bryan 1966. (Reference is occasionally made, as well, to Delafosse 1897, a work of limited value due largely to the source of the material, which was gathered by prompting a speaker of Sara who had forgotten most of his native language with Bagirmi forms. Delafosse's material occasionally provides, however, confirmation of certain points by means of early forms which represent 'Bagirmiisms'.) Comparative material for the Eastern languages (Bongo, Baka etc.) and for the Central languages (Kresh, Yulu, Kara etc.) has been drawn from Santandrea 1963, and from the material in Tucker and Bryan 1966.

1. Introductory

Certain issues involving nasality in the Sara languages pose problems of analysis, both synchronic and diachronic, which we shall examine under two headings:

1.1. Nasalized vowels. All of the languages present, descriptively, surface nasal vowels which appear to function distinctively in underlying representations. Previous analyses, however, have called into question the synchronic reality of such underlying units, based, in large part, on claims about the diachronic source(s) of the nasal vowels

of the Sara languages. In section 2, we shall examine these claims and reject them, on the basis of comparative evidence. Given the historical analysis, we shall discuss the development of the synchronic situation, and consider questions of synchronic analysis, in the context of more general issues involving nasalization.

1.2. Nasal stems. A related, but more complex set of problems revolves around certain morphemes which, on the surface, are largely or totally nasalized. In section 3, we present comparative evidence which mirrors the situation among the nasal vowels, and examine possible directions toward a solution, in the light of the discussion in section 2.

2. Nasalized vowels

2.1. The following nasal vowel phonemes are reported for the Sara languages:

(1) Sa. (Hallaire and Robinne 1959:xviii): *(ĩ)*[1] *ɛ̃*[2] *ã õ* (cf. Delafosse 1897:19, *ẽ ã õ*)

(2) Mb. (Fortier 1961:xi): *ĩ ẽ ã õ*
Mb. (Caprile 1968:18ff.): *ĩ ẽ ã õ (ũ)*[3]

(3) Ng. (Vandame 1963:14–15, 29ff.): *ĩ ẽ ã ɔ̃*[2]
Ng. (Thayer 1971:1–2, 6ff.): *ĩ ẽ ɔ̃*[4] *ã õ (ũ)*[3].

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[1] The existence of */ĩ/* in Sa. is problematic. While Hallaire and Robinne describe only the nasalized glide */ỹ/* (word-final position only), several items with tone-bearing *[ỹ]* may well represent */ĩ/*; Hallaire and Robinne record, in fact, the following items with explicit ⟨*ĩ*⟩: *mâĩ̀* to separate (adversaries), *mâĩ́* venom, *mâĩ̂* attention, cf. *màỹ* stutteringly. We return to Sa. */ĩ/* briefly in 2.4. (Compare Mb. */ũ/*, n. 3 below.)

[2] The phonological status of *[ɛ]* vs. *[e]* and *[ɔ]* vs. *[o]* is controversial for all of the Sara languages, and probably involves a restricted [+ ATR] harmony, although the details are at present unclear. The problem, however, has no bearing on any of the issues discussed here, and will be ignored in the subsequent discussion.

[3] Caprile (1968:22) reports three possible cases of */ũ/*, each one problematic, and the total not sufficient to establish */ũ/*, although such an underlying segment may be in the process of developing; cf. n. 13 below for one of the cases. The only case of *[ũ]* in Thayer's corpus is *pùû̀* 'itch in the throat', which is most likely an expressive formation; cf. Sa. *pù* onomat.: describes the sound of breathing, *pí* onomat.: describes the sound of deep breath, and further, e.g. *pâ* onomat.: describes a dry, cracking sound, *pà* onomat.: describes a bursting sound (cf. French *pan* bang! etc.).

[4] The synchronic status of */ɔ/* is again controversial for all the languages, but has no bearing on the present discussion.

The grammars support the 'phonemic' status of the nasal vowels with reference to the high frequency, in all the languages, of words which contrast minimally with respect to nasality, all other things being equal, including tone (lexical H/M/L for all the languages), e.g.:

(4) Sa.	*kṍ*	mother	cf.	*kó*	grain
	tɛ̃́	to open		*tɛ̀*	to exit
Mb.	*dṍ*	to bite		*dò*	wound
	bɨ̃́	hair		*bì*	mortar
Ng.	*sɨ̃́*	excrement		*sì*	to be seated
	mɑ̃́	to plant		*mà*	I, me.

An important distributional fact is that such distinctive contrasts occur only in word-final position. On the other hand, the facts of contextually determined vowel nasalization are as follows: underlying oral vowels are strongly nasalized preceding prenasalized stops (*/ᵐb/*, */ⁿd/* etc.), and optionally receive light nasalization preceding the nasal consonants */m/*, */n/*. Finally, the nasalized vowels participate in no morphophonemic alternations of any sort (such as with oral vowel plus nasal consonant, etc.).

2.2. The phonological descriptions of the Sara languages have elicited the following evaluations: for Sa., Bouquiaux 1964; for Mb., Tubiana 1962; for Ng., Thomas 1965. All of these critiques consider the synchronic status of the nasal vowels to be questionable a priori, and prefer to remove them from underlying (and perhaps surface) representations in favor of sequences of oral vowel plus some nasal consonant.

Thomas (1965:245) suggests that the nasal vowels may simply reflect oral vowels followed by a final retroflex or palatal nasal, misheard (by European ears) as [-Ṽ#]; similarly already Bouquiaux (1964:267): 'il est assez vraisemblable qu'une nasale, aussi peu familière à des oreilles européennes que la rétroflexe n'ait été perçue que par la nasalisation concomitante de la voyelle précédente'. Tubiana offers a similar interpretation (1962:337–8), but opts for a sequence oral vowel plus velar nasal, given the fact that in Mb., *[ŋ]* in final position is 'barely audible' (Fortier 1961:ix).

These analyses, however, are untenable. Recall that nasalization before nasal consonants (other than prenasalized stops) is light, if it appears at all. The distinctively nasalized vowels, however, are universally described as strongly nasal. These facts cannot be reconciled. (Cf. also the derivational and cross-language facts concerning sequences -VN(+)C-, where N is a nasal consonant. In such cases,

derivational forms either retain the nasal, or delete it, with no trace of concomitant nasalization: thus Mb. */ngɔn/* 'child' + */kõ/* 'mother' → *[ngɔnkõ]* 'brother, sister', but Sa. (and Ng.) */ngón+kṍ/* → *[ngòkṍ]*; cf. Ng. *mùntá* 'three', but Sa. *mùtá*, Mb. *mita*.)

Further, the choice of nasal consonant — palatal, retroflex, or velar — is essentially arbitrary. One could argue for the palatal nasal on the grounds that it represents a case vide (cf. */ⁿj/*), while *[ŋ]* appears otherwise only as the surface realization of the prenasalized stop */ᵑg/* in word-final position (cf. Mb. *[boŋ]* 'tail', *[boᵑga]* 'his tail'), and a retroflex nasal is purely hypothetical[5]. Finally, the absence of any morphophonemic alternations [Ṽ] ≈ [VN] (which might be expected, for example, with vowel-initial suffixes) supports the conclusion that there is no synchronic evidence for a sequence vowel plus nasal consonant.

2.3. From a diachronic point of view, however, routine procedures of reconstruction call for the derivation of nasalized vowels from precisely such sequences (exclusive of so-called 'spontaneous nasalization', to which we return below.) In fact, comparative evidence from outside the Sara languages allows for the reconstruction of a single source for the nasal vowels of the Sara languages, namely *-V*ny*(V) (where *ny* represents a palatal nasal). Consider the following comparisons:

(5) white, brilliant[6]
Sa. *kaɲ́*, Ng. (V.) *káĩ*[7], Mb. *kɛ̃* whiteness
cf. Bongo *ko(i)ny*, Baka *bi-kɛnyɛ*, Yulu *kànyì*

(6) run, flee
Sa. *àɲ́*, Ng. (V.) *àĩ*[7]/(Th.) *àɨ̃*[7], Mb. *aĩ*
cf. Bagirmi *an*^{la8}, Kara *anya*

---

[5] The slightly retroflex character of certain segments, e.g. */l/*, */r/*, */d/*, does not constitute a sufficient basis for a case vide argument. For further discussion of the possibiltiy of a retroflex nasal, see section 3 below.

[6] All cognates share the head meaning(s) unless otherwise specified.

[7] According to Vandame's description, pre- and post-vocalic high vowels optionally (and generally) desyllabify to glides, while such vowels are generally syllabic in Thayer's material.

[8] Gaden describes a series of 'sons mouillés', notated *n*', *d*' etc.: it is unclear whether Gaden refers to palatalized or palatal segments — although corresponding forms in Tucker and Bryan's more recent material are palatals (*[ɲ, ǰ]* etc.), we retain Gaden's 'palatalized' notation. Raised vowels refer to so-called 'semi-mute' vowels (cf. Delafosse 1897:19, Gaden 1909:8–10, Tucker and Bryan

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(7) mother
Sa. *kɔ́*, Ng. (V.) *kɔ́*/(Th.) *kɔ́*, Mb. *kɔ̃*
cf. Bagirmi *kun'u*, Kara *kunyi*

(8) leave, abandon, stop
Sa. *la̰*, Ng. (V.) *ia̰*[7]/(Th.) *ya̰*[7], Mb. *(i)yà*
cf. Bagirmi *in'a*, Kara *ənya*, Bongo *anya*

(9) goat
Sa. *bia̰*, Ng. (V.) *bia̰*/(Th.) *bía̰*, Mb. *ɓia̰*/(Cap. 19) *bìà*
cf. Bagirmi *bin'a*, Kara *vənynya*, Bongo *bìnyà* (Tucker and Bryan 1966:79), Jur-Beli *binya*

(10) bite, prick
Sa. *dɔ́*, Ng. (V.) *dɔ̀*, Mb. *dɔ̃*
cf. Bagirmi *d'on'o* tickle, bite, Kara *djonyo*[9]

(11) excrement
Ng. (V.) *sɨ̀*[10]
cf. Bagirmi *sin'i* fosse à ordures creusée dans les cases et servant de cabinets

(12) eat (delicately)
Sa. *ɔ́*, Ng. (V., Th.) *ɔ̀*, Mb. *ɔ̃*
cf. Kara *onyi* lick[11], Bongo *ony'* (Tucker and Bryan 1966:77)

(13) come from
Sa. *ì*, Ng. (V.) *ɨ̀*/(Th.)*ɨ́*
cf. Bagirmi *in'*.

1966:4,60–1), sometimes also referred to as 'shadow vowels' or 'vowel-colored breaths'. Such vowels differ from normally articulated vowels in linguistically significant ways throughout the Bongo-Bagirmi languages, both synchronically and diachronically. We shall briefly examine a diachronic distinction below; cf. n. 12.

[9] For Bagirmi *d'*-, Kara *dj-* ≈ Sara et al. *d-*, cf. head: Sa. *dɔ̀*, Ng. (V., Th.) *dɔ̀*, Mb. *dɔ*, Bongo *do*, Baka *dɔ̀*, Jur-Beli *dɔ*, but Bagirmi *d'o*, Kara *djú*. (The process is more general: cf. Bongo *hedi*, Kara *kédjì* urine; Yulu *ndiisa*, Kara *ndjèsá* broom.)

[10] A cognate is not quotable from Sa., which resorts to periphrasis (*yede*, derived from the verb 'to fall'.) The inherited word is found in Mb., cf. (17) below.

[11] The semantics are satisfactory: Sa. *ɔ́* etc. is used only for substances 'which do not require much effort in chewing' (cf. Hallaire and Robinne 1959:273), especially sauces and stews. (A different verb is used for meat, etc.) The original meaning is perhaps 'lick'.

There is good evidence, then, that at least some of the nasalized vowels derive from earlier sequences *-V*ny*, *-V*ny*V, or *-V*ny*^{V}[12]. With regard to other possible sequences of vowel plus nasal consonant, there is almost no evidence to suggest that any of them results in synchronic nasalized vowel[13]. As a result of general processes which delete original final vowels (and semi-mute vowels more readily than others), original sequences *-*n*V/-*n*^{V}, *-*m*V/-*m*^{V}, or *-*ŋ*V/-*ŋ*^{V} appear simply as -*n*(V), -*m*(V), and -*ŋ*(V) in the Sara languages.

A surprising fact, however, remains: many of the words with nasal vowel show no trace of earlier *-*ny*, or any other nasal segment. In short, such words have, with respect to nasality, no 'natural' diachronic source, as shown by the following comparative data:

(14) flesh; (wild) animal
Sa. *dɑ̰* (*dɑ̃*, Tucker and Bryan 1966:60), Ng. (V.) *dɑ̀*/(Th.) *dɑ́*, Mb. (Cap. 28) *dɑ̰*
cf. Bagirmi *d'a (ja*, Tucker and Bryan 1966:78, cf. Del. *za)*, Jur-Beli *dɑ́*

(15) (body) hair
Sa. *bɨ̀*, Ng. (V.) *bɨ̀*/(Th.) *b* , Mb. (Cap. 21) *bɨ̀*/(Dict. xi) *bɨ̃*[14]
cf. Bagirmi *bi* (Del. *bi*), Baka *bí*, Yulu *íbì*, Bongo *bɨ̀*, Jur-Beli *ɓi*[15]

(16) drink
Sa. *áỹ*, Ng. (V.) *aì*/(Th.) *àì*, Mb. (Cap. 19) *àɨ̀*
cf. Bagirmi *áy*^{ó} (Tucker and Bryan 1966:77, cf. Gaden *ay*^{o}, Del. *ay*^{e}), Yulu *ɔ́yo*, Bongo *aye*, Kara *áyà*, Jur-Beli *iɛ*

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[12] Given differing developments, certain details remain to be clarified. It seems reasonable to suggest three treatments: (a) *-V*ny*$^{(V)}$ → -Ṽ, as in items (7), (10), (11), (12); (b) *-*inya* → -*i(y)ã*, as in items (8) and (9); and (c) *-*any*(V) → -*aĩ*, where -*ĩ* may he non-syllabic, cf. (5) and (6). The chronology of these processes is not unitary: (b) is relatively late, judging from Sa. *binyà* still in Tucker and Bryan's material (1966:68, cf. Del. *binya*) and the fact that synchronically, in Ng., *ny*V is optionally realized as *y*Ṽ (Thayer 1971:12). We return to similar facts from Sa. and Mb. below, in 3.4.

[13] A possible case is winged termite (black): Sa. *wɑ̀*, Ng. (V.) *uɑ̀*, Mb. *wɑ̃*, cf. Baka *ùŋà*, Yulu *óŋà*. But the status of -*ŋ*- in the Baka and Yulu forms is not certain, since at least Yulu *ŋ* alternates with *ny* in other forms, cf. scorpion, Bongo *hiny*, Jur-Beli *hinya*, Kara *senyi*, but Yulu *siŋ(ŋ)i*, which probably belong with Ng. (V.) *ɨ̀*; bite, Kara *djonyo* etc. (item (10) above), but Yulu *juŋ(ŋ)u*. Such a process may be relatively recent, judging from e.g. Mb. (Cap. 22) *ũ̀* 'to pour', cf. Sa. *òŋ* id., an archaic verb found only in certain fixed expressions.

[14] But also *ɓii*, Dict. 14.

[15] Cf. also Kresh *bib(i)-ušu* beard, Tucker and Bryan 1966:82.

(17) excrement
Mb. (Cap.) *kɨ̀*/(Dict.) *ki*
cf. Baka *ci*, Yulu *kìí*, Bongo *cɨ́*, Kara *cì*

(18) illness
Sa. *mwɛ̃ỹ*, Ng. (V.) *mòi*, Mb. (Cap. 19) *mɔ̀ɨ̀*/(Dict.) *moy*
cf. Bagirmi *muey°*

(19) tongue
Sa. *ndɔ̀* (cf. Del. *ndal[a]*), Ng. (Th.) *ndɔ̀*, Mb. *ndõ*
cf. Jur-Beli *ndɔ*[16].

The salient fact about these forms is that the nasal vowel is not consistent among Sa., Ng. and Mb., whereas the items with nasal etymologies have consistent nasal vowel. If one examines more closely the stock of words with nasal vowel, there is in fact a sizeable group of forms which vary from language to language (and from dialect to dialect) with respect to nasal versus oral vowel. Although comparative material is often lacking for such forms, the existing material, as well as the distribution of forms like those in (14)–(19) versus (5)–(13), suggest that such items stem from non-nasal sources. Consider the following examples:

(20) Sa. *nà* moon, month, festival, Ng. (V.) *nài*/(Th.) *naɨ̀*, Mb. *nâ* (Cap., Dict.); cf. Bagirmi *nap* (Del. *nahU*)
(21) Sa. *mɛ̀hɛ̃* six, Ng. (V.) *mìsá*/(Th.) *mìsâ;* cf. Bagirmi *miga*
(22) Sa. *líâ* chameleon, Ng. (V.) *líà*, Mb. *ki-loĩ*
(23) Sa. *Riã* to tear, Ng. (V.) *rià*, Mb. *ria*
(24) Sa. *míà* throwing knife, Ng. (V.) *mìà*, Mb. *mia*
(25) Sa. *tâ* to stir; sauce, Ng. (V.) *tá* to prepare sauce, Mb. *tã* sauce
(26) Sa. *tõ* to bind, Ng. (V., Th.) *tó*
(27) Sa. *ndɛ̃́* to drip, Ng. (V.) *ndèi*, Mb. *nde*
(28) Ng. (Th.) *sɨ̂* to eat without sauce, Mb. *sii* (Dict. 189) adv.: without sauce
(29) Ng. (V.) *sɨ́* morning, Mb. *sí* adv.: in the morning
(30) Ng. (V.) *nài* to remain, (Th.) *naɨ̂*
(31) Ng. (V.) *tí* to detach, (Th.) *tɨ́* to tear off.

We can determine, then, that the nasal vowels of the Sara languages have two 'sources': a nasal source, from original sequences involving a palatal nasal, resulting in nasalized vowel in all Sara languages; and a 'non-nasal' source — for these cases, there is no apparent

[16] Cf. voice, word, language: Sa. *ndù*, Ng. (V., Th.) *ndù*, Bongo *ndu*, Kara *nduju*.

explanation for vowel nasality, and the synchronic distribution of nasal vowels among the Sara languages is inconsistent.

2.4. It is a commonplace assumption, as we have noted, that nasalized vowels result, in one way or another, from the loss of a nasal consonant (cf. Ferguson's universal statement, 1966:59). Real cases of 'spontaneous nasalization', on the other hand, are said to be 'all but unheard of' (Chen 1973:227); Ferguson (loc. cit.) records only a single case (Iroquoian), apart from the possibilities of borrowing and 'analogy'. Instances of borrowing include Algonquian (from Iroquoian), and perhaps Bengali (see Ruhlen 1973:9 for references)[17]. Possible 'real' cases, in addition to Iroquoian, include Panamanian Spanish, South Ethiopic languages (e.g. Amharic), Tibetan, Central Thai, and Greenlandic (see Schourup 1973:214 n. 2 and Ruhlen 1973:9–10 for references[18]).

As for 'analogy', Ruhlen (loc. cit.) records a single example, from German dialects in which e.g. 'Floh' is realized *[flõ:]*, based on realizations like *[zõ:]* (beside *[zõ:n]*) for e.g. 'Sohn'[19].

How, then, do we account for the apparent secondary development of non-etymological nasal vowels in the Sara languages? The explanation which imposes itself is that the secondary nasal vowels result, somehow, from an extension of 'phonemic' nasality to etymologically non-nasal forms. Unlike the German case just mentioned, we have to do with a widespread process, which must have occurred, moreover, soon after the development of distinctive nasality among the vowels.

It is theoretically possible, though scarcely probable, that an early 'denasalizing' process external to the Sara languages removed nasal segments from many words, with the inconsistent Sa./Ng./Mb. nasal vowels representing relics of such earlier forms. But there is no comparative evidence whatsoever that could support this or similar theories designed to provide 'natural' sources for the stock of words with nasal vowel but no apparent nasal source.

If, then, distinctive nasality was secondarily exploited by etymologically non-nasal forms, it is desirable to delimit the factors which may have contributed to such a development.

[17] The classification of the cases mentioned by Chen is not clear; see Chen 1973:227 n. 5 for references (Chinese (South Min dialects), French (Swiss French dialects), Japanese (Tohoku dialects; anecdotal information only)).

[18] A case deserving further investigation (i.e. comparison with Uto-Aztecan and Tanoan) concerns the nasal vowels reconstructed for Proto-Tiwa (and Proto-Tanoan?) by Trager, 1946:221.

[19] The interpretation rests, at least in part, on rhymes in Goethe.

Consider the possibilities for phonological contrast among the Sara languages. All of the languages present lexical H/M/L tonal contrasts, as mentioned above. Further, there exist neither contour tones, nor distinctive vowel (or consonant) length. Exclusive of nasality, then, distinctive contrasts among (e.g. CV) morphemes are limited to tone, other things being equal. With the appearance of distinctive nasalization, however, possibilities for contrast are doubled. We suggest, then, that with the appearance of etymological, phonologically distinctive vowel nasality, the languages resorted to secondary marking of morphemes by means of (non-etymological) nasality in order to maintain or produce phonological contrasts.

Although it is difficult to test such a hypothesis, certain descriptive facts offer some support. First, by far the majority of surface nasal vowels appear precisely in CṼ morphemes, as opposed to the other two common structures in which nasal vowels may appear, namely VCṼ and CVCṼ. Second, judging from the abundant lexical material of Hallaire and Robinne 1959, perfect homophones (including tone) are surprisingly frequent in general, and particularly frequent among CV morphemes, where distinctions of vowel nasality represent the unmarked situation. Third, the skewed attestations of secondarily nasal vowels indicate both that different languages organize sets of phonological oppositions among morphemes differently, and also that the secondary nasalizations represent a productive process, and appear to have spread in time to more and more forms. Thus, with regard to the first point, a pair such as Ng. *tá* 'to prepare sauce' vs. Sa. *tá̃* (cf. (25) above) is to be understood in the context of a set of synchronic oppositions:

(32) Ng.	*tà*	to take	vs.	Sa.	*tà*
	tà	mouth, end			*tà*
	tá	to prepare sauce			*tá̃*
	tă	to split open			*tẵ.*

With regard to the second point, items like (20), (21), (30) and (31) indicate that the Moundou dialect of Ng. has undergone, over roughly a twelve-year period, a steady increase in the number of secondarily marked nasal vowels[20].

Several questions remain unanswered, from a historical point of view. Why, for example, do some items with non-nasal etymology show

[20] The field work for Vandame's description was completed in 1957 (Vandame 1963:x).

consistent nasalization in all (or most) languages (e.g. hair, item (15) above), while the distribution, for others, appears quite haphazard? One automatically hypothesizes chronological distinctions in such cases; but the material presently available is insufficient to determine relative chronologies with much certainty.

Item (15) raises, in addition, the problem of Sa. /ĩ/ (cf. n. 1, and also Sa. *ì* in (13)): while /ĩ/ in Sa. is clearly on the rise (and also perhaps Mb. /ũ/, cf. n. 3), its development has lagged far behind Ng. and Mb. /ĩ/, though Sa., if anything, shows a higher proportion of non-etymological nasal vowels, in general, than Ng. and Mb. (A possible source of illumination would be the proposed universal relationships between the spread of nasalized vowels and vowel height. Unfortunately, however, the existing claims are contradictory, and the correlations are poorly understood. See Chen 1973 and Ruhlen 1973 for some discussion.)

A final question concerns the synchronic analysis of the nasalized vowels of the Sara languages. The structuralist orientation of the critiques by Thomas, Tubiana and Bouquiaux, with which we began, virtually demanded the rejection of 'phonemic' nasal vowels. Ruhlen argues (1973:18ff.) that early generative theory inherited this bias, and blindly called for underlying representations of oral vowel plus nasal consonant in some cases which may not have been justified. More recently, as Ruhlen notes, one could appeal to principles such as Kiparsky's well-known Alternation Condition, according to which nasal vowels which do not present morphophonemic alternations with oral vowel plus nasal consonant could receive underlying [+ nasal] specification.

There exist, as we have mentioned, no such morphophonemic alternations. Further, although the diachronic explanation presented above concerning secondarily nasalized vowels implies a nearly 'prosodic' function of nasality, there are no synchronic processes (assimilation, spreading etc.) one could associate with a suprasegmental (or autosegmental) treatment of nasality. The Sara situation, then, is fundamentally different from the systems of nasally specified morphemes in e.g. Desano (Kaye 1971) or Guarani (Goldsmith 1976).

There is no particular reason, finally, to treat etymological nasal vowels differently from non-etymological ones: synchronically, they are functionally identical. We conclude, then, that regardless of origin, the nasal vowels must be specified as such in underlying phonological representations.

3. Nasal stems

3.1. Hallaire and Robinne (1959:xx) offer the following description for a group of irregular Sa. forms:

> Une vingtaine (ou un peu plus) de monèmes sont souvent entendus entièrement nasalisés sauf peut-être l'occlusive initiale, mais certainement la consonne interne, qui est toujours un */r/*.

Precisely the same phenomenon is described for Ng. and Mb. (cf. Vandame 1963:9,33, Fortier 1961:xii, and Caprile 1968:20.)

3.2. Essentially two explanations have been offered for such forms: one, that *[-r̃-]* derives from a (probably retroflex) prenasalized segment */ⁿr/* (Bouquiaux 1964:265, Thomas 1965:245, Caprile 1968:30); or alternatively (Tubiana 1962:338) that *[-r̃-]* simply represents a retroflex nasal. For Bouquiaux and Thomas, as well as for Tucker and Bryan (1966:62), prenasalized */ⁿr/* is an areal feature of sorts, to be compared principally with Zande */ⁿr/*, which is generally realized as a nasalized flap *[r̃]*. (For Tubiana, once again, part of the attraction of a retroflex nasal is its strangeness to European ears.)

3.3. In view of widespread surface confusions among *[d]*, *[r]* and *[l]* (all slightly retroflex), the hypothesis of a retroflex nasal segment of some sort is perhaps well-motivated. But these explanations meet with serious difficulties. A prenasalized segment can admittedly explain heavy word-internal nasalization, given the general properties of prenasalized stops. But the word-final nasalization, explained by Tubiana et al. in the same way as for the other nasal vowels discussed above, remains problematic. A few cases, perhaps, allow for an explanation in terms of final nasal consonant: thus for e.g. Sa. *hárá* 'with hanging tongue (dogs)', a dialectal form *harang* is recorded (and cf. the surface contrast with *hárá* 'thorny; acacia siberiana'); Sa. *ngórông*/Mb. *ngorong* 'sickle' (cf. Sa. *ngórông* 'laziness'). Otherwise, one is forced to assume that all V-final words happen, purely coincidentally, to have underlying word-final nasal vowels in addition to internal nasalization — a possibility which cannot be excluded, but which is hardly attractive, and by no means imposing.

A further difficulty concerns the comparison with Zande. It is quite clear that Zande surface realizations of */ⁿr/* attest no nasalization preceding *[r̃]*, but strong nasalization following it: Gore (1931a, b), who was evidently unaware of the possibility of analyzing *[r̃]* as */ⁿr/* (for which see Tucker and Bryan 1966:143), records no cases of sequences *Ṽr*, but literally hundreds with *rṼ*, clearly representing *[r̃Ṽ]*, since syllable boundary invariably precedes the */r/*, and the

entire syllable is described as nasalized (Gore 1931a:5, 1931b:vi)[21]. It is extremely difficult, then, to accept the explanation that the internal nasalization of the Sa. etc. words reflects Zande influence, or a similar 'areal' characteristic[22].

Turning to the data, we find the following distribution. Some forms appear to show nasalization in two[23] of the languages:

(33) Sa. *mbárá* lump of honey, Mb. *mbãr*

(34) Sa. *yárá* liane, Mb. *yãr*

(35) Sa. *yérè* multicolored, spotted, Mb. *yẽr̃ẽ*[24].

In general, however, nasalization is not consistent:

(36) Sa. *ndárỹá* centipede, Mb. *ndari*, Ng. *ndàrè*

(37) Sa. *kárá* batrachian, Mb. *kãr* kind of toad, Ng. *kár* toad

(38) Sa. *dúrú* kalao (bird), Mb. *duru*

(39) Ng. *ndúrí* twisted, Mb. *ndui*, Sa. *ndiu*

(40) Sa. *ngèrwẽ̀* type of large lizard, Mb. *ngarway*.

A slightly different situation appears in (41):

(41) Sa. *gárá* arrow, Mb. (Cap.) *kàrã*/(Dict.) *kerã*.

Although comparative material outside the Sara languages is meager, the existing evidence points to non-nasal etymologies; with (37), cf. Bagirmi *kar*^{a} toad, and with (41) cf. Bongo *kire*, Baka *kéré*, Kara *kər*.

3.4. We seem to find, then, essentially the same situation as for the word-final nasal vowels: some (if not all) forms clearly contain non-etymological nasalization, versus completely non-nasal forms elsewhere. The problem of accounting for the internal nasal vowels, however, remains: one could easily posit, say, 'nasally marked' morphemes, allowing for spreading properties which would exclude, in most cases, the initial consonant. Such an analysis may well account for the synchronic situation, and could be supported, again, by the

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[21] Only two exceptions are found in Gore's dictionary, and these may admit of other explanations: *ẽrehere* smooth, glossy, cf. *ũrehure* mane-like.

[22] No satisfactory analogues exist, to my knowledge, throughout the linguistic area described by Tucker and Bryan. The nasalized vowels of the Teda-Tubu dialects (1966:169) result regularly from the loss of nasal consonants. Although nasalized vowels are said to be 'common' in Banda-Gbaya-Ngbandi (1966:85), the Banda nasalized vowels occur only before prenasalized stops (Tisserant 1931:20–1). No information is available regarding the nasalized vowels of Mba (Tucker and Bryan 1966:108) or of Koma and Ganza (1966:356).

[23] Unfortunately, very few forms are quotable from Ng.

[24] The initial *y-* in these forms may also be nasalized; cf. the discussion in 3.4 below.

relatively high number of contrasts involving nasality. But some further insight into the development of such a system may be achieved by reference to a hitherto misunderstood factor.

Bouquiaux (1964:267) noticed the following four apparently exceptional cases in Sa., with nasal vowel and word-final (non-nasal) /r/ or /l/:

(42) Sa. *yə̃r* to dazzle *niõl* to nauseate
yə̃l to be weak *niẽr* to be dirty.

Bouquiaux constructs an argument, based on these forms, for a retroflex nasal:

> Comme rien dans ce cas ne justifierait la nasalité de la voyelle, il faudrait en déduire que c'est la consonne finale qui est une nasale, la nasalité de la voyelle étant concomitante et non significative . . . il semble que [Hallaire et Robinne] aient perçu à la fois la nasalité et le point d'articulation rétroflexe.

This argument is entirely specious, since there is, in fact, independent justification for the nasal vowels of these forms based on the behavior of /*ny*/ in the Sara languages. As we have seen, earlier combinations involving final **-ny* (V, ^{V}) result either in nasalized vowel or in what is described, in Sa., as a nasalized glide *[ỹ]*. Recall, in addition (n. 12), that in Ng., /*ny*/ is optionally realized as [*y*Ṽ]; similarly, in Mb., initial *y-* is described as nasalized exactly in the cases we have considered, namely before nasal vowel (e.g. *(i)yã* as *[ỹã]* < **inya*, (8) above) or as the initial segment of nasal stems (*yãr̃* 'liane' as *[ỹãr̃]*, Fortier 1961:xi). For Sa., Hallaire and Robinne do not set up a segment /*ny*/; but consider their remark (1959:386) that 'beaucoup de mots commençant par /*ni*/ pourraient s'écrire avec un /*y*/ initial', thus e.g. *nià̃r̃á* 'liane' = *yà̃r̃á*. There is, then, ample evidence that Sa. *ni*V-/*y*Ṽ- represents /*ny*V-/, and therefore Sa. simply shows the same treatment of this sequence as Ng. and Mb., a treatment to be expected in any case on the basis of the historical developments of that sequence already demonstrated for all of the languages. Thus the nasal vowels in the forms in (42) are easily accounted for by the sequence /*ny*V-/, and have nothing whatsoever to do with a hypothetical word-final retroflex nasal.

There exists, then, a well-motivated source for surface word-internal nasal vowels. Given a form /*ny*V*r*V/: in order to allow for the possibility of spreading (though not necessarily to account for it), one could think of e.g. Schourup's constraint (1973:200) that prevents nasalization from spreading through obstruents, but which allows

progressive spreading through liquids, which is not infrequent (cf. the non-prosodic cases of spreading, 197ff.). Once phonetic spreading took place in forms like (*yãra* →) *yãr̃ã* 'liane', (*yẽre* →) *yẽr̃ẽ* 'spotted', the possibility of secondary nasal marking could easily have arisen, leading to contrasts such as Sa. *kã́r̃á* 'pimento', *kárà* 'rust', etc.

Consider, in addition, the virtual non-existence of forms of the structure CV*r*Ṽ (cf. Sa. *hárà* 'thorny; acacia siberiana', cited above in 3.3, and the Mb. forms in (41)). This fact allows us to assume underlying (or historical) forms of the shape CV*r*Ṽ for some (or even all) of the forms in (33)–(41), with -Ṽ# either etymological (cf. *hã́r̃à: harang* discussed above, and n. 13), or non-etymological, as is clearly the case in (41). We may then account for the nasality of the first vowel by positing a regressive nasal spreading rule, i.e. the mirror image of the process envisioned above for cases like *yãr̃ã*. If, in fact, as seems to be the case, regressive nasal spreading is more frequent than progressive spreading, then the regressive analysis may be the more natural one[25], although the progressive analysis remains available for at least a few cases.

Several problems remain, however. The restriction to forms with medial /*r*/ is not easily accounted for. A clue is perhaps afforded by the Ng. optional realization *[-n-]*[26], as opposed to *[-r̃-]*, but one is reduced to ad hoc speculation as to possible developments[27]. One wonders, in addition, whether the identical vowel quality of the Sa. forms, generally CṼ$_i$r̃Ṽ$_i$ (versus Mb. CṼr̃), has any bearing on the analysis[28].

A final point concerns the productivity of whatever process is responsible for the secondary nasality of many (if not all) of these forms. Unlike the process described in section 2, which shows clear signs of productivity, these forms represent, numerically, an extremely limited class. Moreover, the most recent information (Thayer 1971) presents no trace of the phenomenon at all: while Vandame records, for Ng., *[tínà]* or *[tĩr̃à]* 'axe', Thayer knows only *[tínà]*, and seems unaware of any such process operative in this or any other forms.

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[25] I am indebted to George N. Clements for suggesting this alternative analysis.

[26] Unambiguously described as apico-dental, and not retroflex.

[27] Dissimilation of e.g. **nyana* → **nyara* (→ **yãra* etc.)??

[28] In addition to the forms in (33)–(41), e.g. *gà̃r̃à* bird, *hṍr̃ô* chalk, *tõ̀r̃ô* type of rat, *kà̃r̃à* spittle, *mbà̃r̃à* trap, etc.

4. Summary

After examining the historical sources of the nasal vowels of the Sara languages, we postulated a process of widespread secondary nasalization. These languages present, then, a case involving the non-etymological spread of nasal vowels far more extensive than any other accounts of 'analogical' nasals in the literature on 'spontaneous nasalization' known to me. The process is productive, lexically based, and provides no support for a synchronic analysis of nasal vowels in terms of oral vowel plus nasal consonant.

The 'nasal stems' of the Sara languages appear to involve essentially the same sort of process; although the details are less clear, the historical and synchronic behavior of the sequence /*ny*V-/ accounts for certain previously misinterpreted forms, and may be important for the diachronic explanation of the nasal stems, which must ultimately appeal to phenomena of nasal spreading. In contrast with the productive secondary marking of nasal vowels, the process responsible for the nasal stems is unproductive, if not extinct.

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Buchbesprechungen

Williamson, Kay (Hrsg.), Benue-Congo Comparative Wordlist. Volume II. From contributions by members of the Benue-Congo Working Group. West African Linguistic Society, University of Ibadan. Ibadan, Nigeria, 1973.

Der zweite Band der Benue-Congo Comparative Wordlist (BCCWL), der die Stichwörter von „man" bis „woman" sowie die Zahlwörter enthält, steht unter der gleichen Zielsetzung wie der erste[1], nämlich lexikalisches Material einer möglichst großen Zahl von Benue-Congo-Sprachen der wissenschaftlichen Öffentlichkeit zugänglich zu machen. Diese Zielsetzung verdient in einer Zeit, in der das simple Veröffentlichen von Primärdaten Seltenheitswert zu bekommen beginnt, besondere Anerkennung.

Der Aufbau und die Präsentation des ersten Bandes sind beibehalten worden: auf je drei Seiten werden die Entsprechungen zu einem der englischen Stichwörter in den verschiedenen BC-Sprachen wiedergegeben, eine vierte Seite enthält Kommentare, meist in Form von tatsächlich rekonstruierten Wurzeln oder von „Pseudorekonstruk-

[1] S. Besprechung des ersten Bandes in AuÜ 53 (1969/70): 290/1 durch A. Vorbichler. Die Bedenken gegenüber der Verwendung der Begriffe „Sprachfamilie" und „Unterfamilie" werden vom gegenwärtigen Rezensenten nicht geteilt.